Problem 1

$\bar{X} = 52, \quad s = 10, \quad n = 100, \quad \mu_0 = 50 \qquad H_0: \mu = 50 \quad H_1: \mu \neq 50$

$$SE = \frac{s}{\sqrt{n}} \Rightarrow SE = \frac{10}{\sqrt{100}}$$

$$= \frac{10}{10}$$

$$= \boxed{1}$$

$$z = \frac{\bar{X} - \mu_0}{SE} \Rightarrow z = \frac{52 - 50}{1}$$

$$= \frac{2}{1}$$

$$= \boxed{2}$$

$\alpha = 0.10$	1.645	Reject H_0
$\alpha = 0.05$	1.96	Reject H_0
$\alpha = 0.01$	2.576	Fail to reject H_0

$CI = \bar{X} \pm 1.96\, SE \Rightarrow CI = 52 \pm 1.96(1)$

$= [52 - 1.96, 52 + 1.96]$

$= [50.04, 53.96]$

Problem 2

$\hat{S} = 3.1, \quad SE = 1.3 \qquad H_0: s = 0 \quad H_1: s \neq 0$

$CI = \hat{S} \pm z\, SE \Rightarrow$ 90% CI $= 3.1 \pm 1.645(1.3)$

$= 3.1 \pm 2.1385$

$= [0.9615, 5.2385]$

95% CI $= 3.1 \pm 1.96(1.3)$

$= 3.1 \pm 2.548$

$= [0.552, 5.648]$

99% CI $= 3.1 \pm 2.576(1.3)$

$= 3.1 \pm 3.3488$

$= [-0.2488,\ 6.4488]$

$$z = \frac{\hat{s} - s_0}{SE} \Rightarrow z = \frac{3.1 - 0}{1.3}$$

$$= 2.385$$

$\alpha = 0.10$	1.645	Reject H_0
$\alpha = 0.05$	1.96	Reject H_0
$\alpha = 0.01$	2.576	Fail to reject H_0

Problem 31

$\bar{X} = 11.5$, $s = 3$, $n = 36$, $\mu_0 = 12$ $\quad H_0 : \mu = 12 \quad H_1 : \mu \neq 12$

$$SE = \frac{s}{\sqrt{2}} \Rightarrow SE = \frac{3}{\sqrt{36}}$$

$$= \frac{3}{6}$$

$$= 0.5$$

$$CI = \bar{X} \pm 1.96(SE) \Rightarrow CI = 11.5 \pm 1.96(0.5)$$

$$= 11.5 \pm 0.98$$

$$= [10.52,\ 12.48]$$

↳ 12 is a plausible value

$$z = \frac{\bar{X} - \mu_0}{SE} \Rightarrow \frac{11.5 - 12}{0.5}$$

$$= -1$$

↳ $|z| < 1.96$

$1 < 1.96 \Rightarrow$ fail to reject H_0

Problem 4

$\bar{X}_A = 25, \quad s_A = 8, \quad n_A = 100$

$\bar{X}_B = 22, \quad s_B = 7, \quad n_B = 100$

$\bar{X}_A - \bar{X}_B \Rightarrow 25 - 22$

$= \boxed{3}$

$SE = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} \Rightarrow SE = \sqrt{\frac{8^2}{100} + \frac{7^2}{100}}$

$= \sqrt{0.64 + 0.49}$

$= \sqrt{1.13}$

$\boxed{SE \approx 1.063}$

$H_0 : \mu_A = \mu_B$

$H_1 : \mu_A \neq \mu_B$

$z = \frac{(\bar{X}_A - \bar{X}_B) - 0}{SE} \Rightarrow z = \frac{25 - 22}{1.063}$

$\boxed{z = 2.82} \Rightarrow |z| > 1.96$, reject null hypothesis

$CI = (\bar{X}_A - \bar{X}_B) \pm 1.96(SE) \Rightarrow CI = 3 \pm 1.96(1.063)$

$= 3 \pm 2.084$

$= [0.916, 5.084]$

Average houses prices differ between the two neighborhoods.

Problem 5

$\bar{X}_A = 14, \quad s_A = 6, \quad n_A = 81$

$\bar{X}_B = 12, \quad s_B = 5, \quad n_B = 64$

$\bar{X}_A - \bar{X}_B \Rightarrow 14 - 12$

$= 2$

$SE = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} \Rightarrow SE = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}}$

$$= \sqrt{\frac{36}{81} + \frac{25}{64}}$$

$$= \sqrt{0.4444 + 0.3906}$$

$$= \sqrt{0.8351}$$

$$\boxed{SE \approx 0.914}$$

$H_0 : \mu_A - \mu_B = 0$

$H_1 : \mu_A - \mu_B \neq 0$

$$z = \frac{(\bar{x}_A - \bar{x}_B) - 0}{SE} \Rightarrow z = \frac{14 - 12}{0.914}$$

$$\boxed{z \approx 2.19} \Rightarrow |z| > 1.96, \text{ we reject the null hypothesis: recovery time differ between the two treatments.}$$

Problem 6

$\bar{x}_A = 3.2, \quad s_A = 1.5, \quad n_A = 49$

$\bar{x}_B = 2.8, \quad s_B = 1.4, \quad n_B = 64$

$$\bar{x}_A - \bar{x}_B = 3.2 - 2.8 = 0.4$$

$$SE = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} \Rightarrow SE = \sqrt{\frac{1.5^2}{49} + \frac{1.4^2}{64}}$$

$$= \sqrt{\frac{2.25}{49} + \frac{1.96}{64}}$$

$$= \sqrt{0.0459 + 0.0306}$$

$$= \sqrt{0.0765}$$

$$\boxed{SE \approx 0.277}$$

$H_0 : \mu_A - \mu_B = 0$

$H_1 : \mu_A - \mu_B \neq 0$

$$z = \frac{(\bar{x}_A - \bar{x}_B) - 0}{SE} \Rightarrow z = \frac{3.2 - 2.8}{0.277}$$

$$\boxed{z \approx 1.45} \Rightarrow |z| \geq 1.96, \text{ fail to reject } H_0, \text{ not enough evidence to conclude a differ in score}$$

$$CI = (\bar{x}_A - \bar{x}_B) \pm 1.96(SE) \Rightarrow CI = 0.4 \pm 1.96(0.277)$$

$$= 0.4 \pm 0.543$$

$$= [-0.143, 0.943]$$

Problems 7 - 10

https://github.com/jonathan-orellana/uu_inference/tree/main/problems